

6 (a) power / heating depends on I^2 M1
 so independent of current direction A1 [2]

(b) *either* maximum power = $I_0^2 R$ or average power = $I_{\text{RMS}}^2 R$ M1
 $I_0 = \sqrt{2} \times I_{\text{RMS}}$ M1
 maximum power = 2 × average power
 ratio = 0.5 A1 [3]

7 (a) force due to E field is equal and opposite to force due to B field R1